2.6.38

AI25BTECH11027 - NAGA BHUVANA

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Question:
If $\mathbf{a} = \hat{i} + \hat{j} + \hat{k}$ and $\mathbf{b} = \hat{j} - \hat{k}$, find a vector $\mathbf{c}$ such that $\mathbf{a} \times \mathbf{c} = \mathbf{b}$ and $\mathbf{a} \cdot \mathbf{c} = 3$

Solution:

$$\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \tag{0.1}$$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \tag{0.2}$$

$$\mathbf{a} \times (\mathbf{a} \times \mathbf{c}) = \mathbf{a} \times \mathbf{b} \tag{0.3}$$

$$(\mathbf{a}^T \mathbf{c})\mathbf{a} - (\|\mathbf{a}\|^2)\mathbf{c} = \mathbf{a} \times \mathbf{b} \tag{0.4}$$

As $\mathbf{a}^T\mathbf{c} = 3$ and $\|\mathbf{a}\|^2 = 3$

$$3\mathbf{a} - 3\mathbf{c} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \tag{0.5}$$

$$3\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 3\mathbf{c} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \tag{0.6}$$

$$3\mathbf{c} = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} \tag{0.7}$$

$$\mathbf{c} = \begin{pmatrix} \frac{5}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix} \tag{0.8}$$

$$\therefore \mathbf{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k} \tag{0.9}$$

Graphical Representation